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MILITARY SPECIFICATION

AMPLIFIER, TWO CHANNEL, INTEGRAL, S-2 COMPASS

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for procurement of one type, two-channel, integral amplifier.

1.2 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-5015	Connectors, Electric, "AN" Type
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing

FSC 6605

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Military (Continued)

MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-19938	Control, Directional, Electrically Driven Gyro, S-2 Compass and S-5 Automatic Pilot
MIL-A-19940	Amplifier, Control, Single Channel, S-2 Compass
MIL-C-19941	Controller, S-2 Compass, Type
MIL-I-21429	Indicator, Compass Repeater, Rotating Dial, Type MW-1
MIL-T-25193	Transmitter, Remote Compass, Type C-2 (Unstabilized)

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment
MS17321	Meter, Time Totalizing Miniature, Digital, 28 Volt DC
MS28053	Meter, Time Totalizing, Digital, 115 Volt 400 Cycle

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Military (Continued)

MS28080 Amplifier, Two Channel, Integral, S-2 Compass and S-5 Automatic Pilot

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Preproduction - This specification makes provision for preproduction testing.

* 3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400.

(b) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5.)

(c) When previously produced models of the amplifier did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

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3.3 Design and Construction - The amplifier shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the amplifier, excluding cables, shall not exceed 5.5 pounds.

* 3.3.2 Reliability -

3.3.2.1 Operational Stability - The amplifier shall operate with satisfactory performance, continuously or intermittently, for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

* 3.3.2.2 Longevity - The amplifier shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The amplifier shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

* 3.3.2.4 Time Totalizing Meter - The amplifier shall contain a time totalizing meter in accordance with MIL-M-7793. The meter shall monitor either the 28V DC power input or the 115V, 400 cps power input. The meter shall conform to MS17321 for 28V or MS28053 for 115V, and shall be readable without removing the amplifier from the case.

3.3.3 Cabling and Connections -

* 3.3.3.1 Cables and Connectors - The amplifier shall provide for the use of cables and connectors in accordance with MIL-E-5400.

* 3.3.3.2 Interconnection Cabling - The amplifier shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the amplifier.

3.3.4 Interchangeability of Reordered Equipment - For reordered amplifiers, interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured amplifier supplied or designated by the procuring activity.

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3.3.5 Interference Control - The generation of radio interference by the amplifier and the vulnerability of the amplifier to radio interference shall be controlled within the limits of MIL-I-6181.

* 3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

* 3.3.7 Nameplates - Nameplate approval for amplifier identification shall be in accordance with MIL-N-18307.

* 3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° C ±5° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 V, 400 ±5 cps and 27.5 ±0.5 V DC

* 3.3.9 Service Conditions - The amplifier shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment.

* 3.3.10 Primary Input Power Requirements - The amplifier shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein.

(a) AC Power (Single Phase) 115V, Category B.

(b) DC Power 28V, Category B.

* 3.3.10.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the amplifier.

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3.3.11 Wiring - The electrical wiring of the amplifier shall be as follows and shall be isolated from the case.

Pin A	-28V DC, ground 115V AC
B	Repeater Excitation, 86.5V 400 cps, out
C	115V AC
D	+28V DC
E, F	$input_1$
F, G	$input_2$
H, I	$output_1$
I, J	$output_2$

* 3.3.12 Standby Provisions - The amplifier shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltage shall not be applied to other parts of the amplifier.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail Requirements -

3.5.1 Amplifier - The amplifier shall be designed to operate from the stator leads of the heading synchro contained in the slaved directional gyro control conforming to MIL-C-19938. The amplifier assembly shall incorporate a quick detachable shockmount base. The design shall be such as to permit service maintenance, repair or overhaul without requiring special tools and fixtures. The amplifier shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to service without failure, shipping, storage and installation.

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3.5.1.1 Output - Each channel of the amplifier shall contain 100 percent degenerative feedback so that the total output voltage is fed back into the input 180 degrees out of phase. Input and output voltages shall be related as specified in the Gain test. Each channel shall provide a power gain of at least 18 decibels. The power output shall be suitable and adequate to position accurately a minimum of 6 rotating card type compass indicators.

* 3.5.1.2 Case - The outline dimensions of the amplifier case shall conform to MS28080. The case shall be finished with a lusterless, black material Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.5.1.3 Electrical Connector - The electrical connector used shall conform to MIL-C-5015 and MS28080. The connector code markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector of the amplifier.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

* 4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests

(b) Initial Production Tests

(c) Acceptance Tests

(d) Longevity Test

* 4.2 Preproduction Tests - Preproduction tests shall be made on an amplifier representative of the production amplifiers to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative

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may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

* 4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the amplifier meets all the requirements of this specification and the contract. As a minimum, preproduction tests shall include all of the tests specified herein under Test Methods.

* 4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production amplifiers shall be delivered prior to the approval of the preproduction sample. Prefabrication of production amplifiers prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of amplifiers to be submitted for acceptance. The preproduction sample shall not be considered as one of the amplifiers under the contract.

* 4.2.4 Production Equipments - Amplifiers supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each amplifier shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for amplifiers already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Initial Production Tests - One of the first ten production amplifiers shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This amplifier shall be selected by the procuring activity after the amplifier has successfully passed the Individual tests. The preproduction sample shall not be selected for this test.

* 4.3.1 Scope of Tests - Each of the amplifiers selected for the Initial Production Tests shall be subjected to the following tests:

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(a)	Sampling Tests	
(b)	Vibration Failure	(4.8.8)
(c)	Radio Noise Interference	(4.8.9)
(d)	High Temperature Exposure	(4.8.10)
(e)	High Altitude - Low Temperature	(4.8.11)
(f)	Humidity	(4.8.12)
(g)	Fungus Resistance	(4.8.13)
(h)	Salt Spray	(4.8.14)

* 4.3.2 Accessory Material - In addition to the complete amplifier submitted for Initial Production Tests, the contractor shall also submit such accessory material and data as are necessary to test the amplifier.

* 4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of amplifiers under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on amplifiers previously accepted when requested by the procuring activity.

* 4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the amplifier shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the amplifier for acceptance.

* 4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

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(a) Individual Tests

(b) Sampling Tests

(c) Reliability Assurance Tests

(d) Special Tests

4.4.1 Individual Tests - Each amplifier submitted for acceptance shall be subjected to the Individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each amplifier accepted shall have passed the following tests:

(a) Examination of Product	(4.8.1)
(b) Dielectric Strength	(4.8.2)
(c) Gain Test	(4.8.3)
(d) Excitation Voltage	(4.8.4)

* 4.4.2 Sampling Tests - Amplifiers selected for sampling tests shall first have passed the Individual tests. Amplifiers shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Amplifiers Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0 1/
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

1/ If by contract action the Initial Production Tests (4.3) are deleted, then a Sample Test shall be conducted on one amplifier from the first 10 produced.

* 4.4.2.1 Scope of Tests - As a minimum, each amplifier selected for sampling tests shall be subjected to the following tests:

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(a) Low Temperature Operation (4.8.5)

(b) High Temperature Operation (4.8.6)

(c) Vibration Error (4.8.7)

* 4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by MIL-STD-781. Tests as required by both the Qualification Phase and Sampling Phase shall be conducted.

* 4.4.3.1 Qualification Phase - Prior to the acceptance of amplifiers under the contract or order, a minimum of three (3) amplifiers shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of amplifiers to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

* 4.4.3.2 Reliability Sampling Phase Tests - Samples of the amplifier shall be tested as outlined in MIL-STD-781 under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level F shall be used. The Accept-Reject critieria for Test Plan II shall be used to determine the length of the tests (until an accept or reject decision is reached).

* 4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those amplifiers used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime, other amplifiers constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of amplifiers at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the amplifier, the amplifier parts, the amplifier quality control, etc., so that the entire contract quantity will show an accept decision.

* 4.4.3.3 Test Details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted to, and approved by, the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following paragraphs shall be considered minimum conditions and shall apply to both the Qualification and Sampling phases.

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* 4.4.3.3.1 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the amplifier shall be operated 2-1/2 hours during each cycle and shall be "OFF" for a period of 1/2 hour.

* 4.4.3.3.2 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.8.

* 4.4.3.3.3 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the tests:

(a) Whenever performance characteristics fall below the acceptance requirement (4.4.3.3.2), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one caused the other parts to fail.

* 4.4.3.3.4 Preventive Maintenance - During the period of the tests, no preventive maintenance may be performed upon the amplifier.

* 4.4.4 Special Tests - Special tests shall be conducted on a quantity of amplifiers for the purpose of checking the effect of any design or material change on the performance of the amplifiers and to assure adequate quality control. The amplifiers selected for special tests may be selected from amplifiers previously subjected to the sampling or reliability assurance tests.

* 4.4.4.1 Special Test Schedule - Selection of amplifiers for special tests shall be made as follows:

(a) On an early amplifier after an engineering or material change.

(b) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

* 4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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* 4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken.

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all amplifiers for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Longevity Test - The longevity test shall be performed in accordance with Test Plan XXVIII of MIL-STD-781 except that 5000 hours operating time shall be performed on one sample of the amplifier. One amplifier used on the Reliability Qualification Plan may be continued until it has accumulated the required 5000 hours. The longevity test procedures shall be similar to those of 4.4.3. After the first 1000 hours on the longevity test other maintenance may be performed provided the cost maximum of servicing and replacement of parts, throughout the longevity test, does not exceed the "cost to repair" specified in the contract. (See 6.2.)

* 4.5.1 Reconditioning of Longevity Test Samples - An amplifier which has been subjected to the longevity test shall be reconditioned as follows:

(a) On completion of the longevity test, the amplifier shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the amplifier for acceptance.

* 4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and longevity tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary

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to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Test Conditions -

* 4.7.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.8.

* 4.7.2 Attitude - Unless otherwise specified, the amplifier shall be tested in its normal operating position.

4.8 Test Methods -

4.8.1 Examination of Product - Each amplifier shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.8.2 Dielectric Strength - A potential of 500 volts, 60 cycles, shall be applied between isolated pins and the case for a period of 5 seconds. The dielectric tester shall be set for one milliampere sensitivity. There shall be no breakdown of insulation or any other permanent damage to the amplifier as a result of this test.

4.8.3 Gain Test - The gain of each channel of the amplifier shall be measured by applying a known voltage to the input of a channel and measuring the output voltage across a load resistor. To avoid tolerance buildup at any specified setting, the same meter shall be used to measure all voltages. The difference in output voltages between two channels in the same amplifier at any of the specified settings shall not exceed 4 percent of the greater reading. The source of the known voltage shall consist of a 1000 ohm potentiometer supplied with 13.5 volts, 400 cycle AC from a transformer. A 400 ohm resistor of adequate wattage shall be used to load the channel being tested. The input voltage, applied across terminals J-20-E and J-20-F, shall be set at the values specified in Table I and the output voltage shall be measured across the load resistor connected between terminals J-20-H and J-20-I. The output voltages shall not exceed the limits specified in Table I. This procedure shall be repeated with the input voltages being applied across the terminals J-20-G and J-20-F and the output voltages measured across J-20-J and J-20-I.

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TABLE I

Input Volts	Output Volts	
Nominal	Minimum	Maximum
0.51	0.46	0.5
2.0	1.84	2.0
4.0	3.68	4.0
8.0	7.36	8.0
10.0	9.20	10.0
13.0	11.96	13.0

4.8.4 Excitation Voltage - The repeater excitation voltage supply shall be measured across a 1000 ohm resistor of adequate wattage connected between terminals J-20-B and J-20-A. The excitation voltage shall be between 83.0 and 90.0 volts.

* 4.8.5 Low Temperature Operation - The amplifier shall be properly connected except that no power shall be applied. The amplifier shall be tested in accordance with Step 2, Temperature and Altitude Tests of MIL-T-5422 for 4 hours. At the end of the 4-hour period and while at the reduced temperature, power shall be applied for 10 minutes. At the end of the 10-minute period and still at the reduced temperature, the amplifier shall be subjected to and shall meet the requirements of the Gain and Excitation Voltage tests.

* 4.8.6 High Temperature Operation - The amplifier shall be properly connected and tested in accordance with Step 6, Temperature and Altitude Tests of MIL-T-5422. At the end of the 4-hour period and while still at the elevated temperature, the amplifier shall be subjected to and shall meet the requirements of the Gain and Excitation Voltage tests.

* 4.8.7 Vibration Error - The amplifier output shall be connected to a compass repeater indicator. The amplifier shall be energized and the reading on the standard indicator shall be noted. The amplifier shall then be vibrated for 5 minutes in accordance with Parts I and II, Vibration Tests, of MIL-T-5422. During the vibration, the reading of the repeater shall not change by more than 2 degrees from the reading prior to the vibration.

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* 4.8.8 Vibration Failure - The amplifier shall be tested in accordance with Parts I and II, Vibration Tests, of MIL-T-5422. At the conclusion of this test, the amplifier shall be subjected to and shall meet the requirements of the Individual tests. There shall be no evidence of damage or irregular operation as a result of the vibration.

4.8.9 Radio Noise Interference - The amplifier shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181.

* 4.8.10 High Temperature Exposure - The amplifier shall be tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422 for a period of 24 hours. The amplifier shall be properly connected but no power shall be applied during the test. During the last 15 minutes of the 24-hour period, power shall be applied. At the end of the 24-hour period and while still at the elevated temperature, the amplifier shall be subjected to and shall meet the requirements of the Gain and Excitation Voltage tests. When the amplifier has been returned to room temperature, the Gain and Excitation Voltage tests shall be repeated.

* 4.8.11 High Altitude-Low Temperature - The amplifier, properly connected but with no power applied, shall be subjected to Step 3, Temperature and Altitude Tests, of MIL-T-5422 for a period of 72 hours. At the end of the exposure period and while still at the temperature and altitude specified, the amplifier shall be subjected to and shall meet the requirements of the Gain and Excitation Voltage tests. After the amplifier has returned to room temperature and pressure it shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.12 Humidity - The amplifier shall be subjected to the Humidity Test of MIL-T-5422 except that 71° C shall be substituted for 50° C in Steps 1 and 2. At the conclusion of the test there shall be no evidence of corrosion. The amplifier shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.13 Fungus Resistance - The amplifier shall be tested in accordance with the Fungus Resistance Test of MIL-T-5422, except that the test period shall be 14 days. At the conclusion of the test, any evidence of the growth of fungi shall be cause for rejection. The equipment shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.14 Salt Spray - The amplifier shall be tested in accordance with the Salt Spray Test of MIL-T-5422 except that the test period shall be 50 hours. At the end of the 50-hour period the amplifier shall be subjected to and shall meet the requirements of the Individual Tests. There shall be no evidence of corrosion or other deterioration which could affect subsequent operation.

* 4.9 Reconditioning of Tested Equipment - Amplifiers which have been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the amplifiers for acceptance.

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* 4.10 Presubmission Testing - No amplifiers shall be submitted by the contractor until they have been tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

* 4.11 Rejection and Retest - Amplifiers which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

* 5.1 General - Each amplifier shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

* 6.1 Intended Use - The amplifier covered by this specification is intended for use as part of the S-2 compass system.

* 6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

(c) Cost to repair (see 4.5).

* 6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

* 6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of the amplifier. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired that their use be investigated. When investigation shows that advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

* 6.5 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

* 6.6 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

* 6.7 Associated Equipment - The amplifier shall operate satisfactorily as part of the S-2 Compass System which includes the following components:

Component	Specification
Amplifier, Control, Single Channel, S-2 Compass	MIL-A-19940

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Component	Specification
Control, Directional, Electrically Driven Gyro, S-2 Compass and S-5 Automatic Pilot	MIL-C-19938
Controller, S-2 Compass, Type	MIL-C-19941
Transmitter, Remote Compass, Type C-2 (Unstabilized)	MIL-T-25193
Indicator, Compass Repeater, Rotating Dial, Type MW-1	MIL-I-21429

* 6.8 Performance Characteristics to be Measured - At least once each week, the amplifier shall be subjected to and shall meet the requirements of the following tests:

(a) Gain Test (see 4.8.3)

(b) Excitation Voltage (see 4.8.4)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-19939A(AS) AMPLIFIER, TWO CHANNEL, INTEGRAL, S-2 COMPASS

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D. C. 20360

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